Out-of-sample results (locked out-of-sample 2022-2025)

Frozen parameters: instrument=Nifty50, k=0.5, window=60, hold=2 days, side=both, cost=0.05% per side.
Run at 2026-11-03 15:49:24.

Reading note, added after the run (5 Nov 2026). Every number below is exactly as the run wrote it. The locked data starts on 3 Jan 2022, but the rule needs the previous 60 overnight gaps to judge what a big gap is, and the run counted them from that first day instead of using the end of 2021. The window was full on 31 Mar 2022, the first signal was possible on 1 Apr and the first entry on 4 Apr 2022, so the first 62 of the 984 days held no position. The effective trading period is April 2022 to December 2025, the CAGR, Sharpe and drawdown below include those flat days, and buy and hold covers the whole of 2022. I cannot say whether the late start helped or hurt, and the test was not rerun. See "Known limitation" in `part_c/README.md` and section 8 of the research note.

1. Data quality of the locked period (same cleaning function, nothing tuned)

rule	series	rows	details
muhurat session (1 hour, holiday)	nifty	3	[datetime.date(2022, 10, 24), datetime.date(2024, 11, 1), datetime.date(2025, 10, 21)]
rows kept after single-series rules	nifty	985	started with 988
muhurat session (1 hour, holiday)	bank	3	[datetime.date(2022, 10, 24), datetime.date(2024, 11, 1), datetime.date(2025, 10, 21)]
rows kept after single-series rules	bank	984	started with 987
date only in nifty (dropped)	nifty	1	[datetime.date(2025, 2, 1)]

Clean table: 984 rows, 2022-01-03 to 2025-12-31.
Days with a move above 5% in either index: 2.

2. In-sample vs out-of-sample, same rule, same parameters

	In-sample 2012-21	Out-of-sample	Buy & hold Nifty, in-sample	Buy & hold Nifty, out-of-sample	Same rule on Bank Nifty, out-of-sample
CAGR %	20.19	9.64	14.4	10.82	-3.54
Ann vol %	15.76	12.11	17.64	14.6	14.74
Sharpe	1.25	0.82	0.85	0.78	-0.17
Max drawdown %	-16.74	-16.38	-37.77	-17.11	-37.7
Days	2444	983	2444	983	983
Trades	661	263	nan	nan	279
Win rate %	51.89	49.05	nan	nan	44.09
Avg win %	1.55	1.3	nan	nan	1.38
Avg loss %	-1.07	-0.96	nan	nan	-1.15

Out-of-sample gross Sharpe (before costs) 1.38, net 0.82. Costs paid: 26.3% of capital. Time in market: 73% of days.

	2022	2023	2024	2025
strategy net %	16.4	1	4.9	16.1
buy & hold %	4.2	19.8	8.8	9.9

Alpha / beta vs always-long Nifty, out-of-sample: average exposure 0.20, beta 0.05, alpha 9.3% a year (t=1.37).

3. Is the underlying anomaly still there? (yesterday's gap -> next tradable day)

period	index	effect bps per 1sd gap	t-stat	n
research 2012-21	Nifty50	12.84	3.91	2442
research 2012-21	BankNifty	18.9	3.05	2442
out-of-sample	Nifty50	-0.18	-0.04	981
out-of-sample	BankNifty	-5.25	-0.88	981

Out-of-sample mean next tradable day return (bps) by gap quintile:

	Q1 (big gap down)	Q2	Q3	Q4	Q5 (big gap up)
bps	-5	6.2	10.3	3.5	4.2

4. Parameter sensitivity out-of-sample (information only, nothing is re-tuned from this)

variant	k	window	hold	net Sharpe	CAGR %	MaxDD %	trades
base	0.5	60	2	0.82	9.64	-16.38	263
k -20%	0.4	60	2	0.44	4.88	-15.66	286
k +20%	0.6	60	2	0.52	5.5	-19.32	250
window -20%	0.5	48	2	0.68	7.95	-21.55	268
window +20%	0.5	72	2	0.66	7.53	-15.05	255
hold -1	0.5	60	1	-0.21	-2.66	-24.26	336
hold +1	0.5	60	3	0.78	9.53	-18.49	243

Cost sensitivity out-of-sample:

cost per side %	net Sharpe	CAGR %
0	1.38	17.29
0.02	1.1	13.4
0.05	0.82	9.64
0.08	0.54	6.01
0.1	0.26	2.49

5. Was the out-of-sample drawdown within what the research-period bootstrap predicted?

Actual out-of-sample max drawdown: -16.4% over 983 days.

- Scenario 1 (edge as measured): median predicted max drawdown -14.9%, worst 5% -24.3%. Share of simulated paths with a drawdown at least as bad as the actual one: 36%.
- Scenario 2 (half the edge): median -19.7%, worst 5% -34.1%. Share at least as bad as the actual one: 72%.

Actual out-of-sample Sharpe 0.82. Share of simulated paths with a Sharpe at least this LOW: scenario 1 16%, scenario 2 66%.

Bootstrap 95% confidence interval for the out-of-sample Sharpe (resampling the out-of-sample days): -0.22 to 1.83 (median 0.82).